

Student 21 **Lakshmi Narayanan E** 192524249RD

5 / 9 submitted

Q1. Banking Transaction System – Exception Handling

CODE

```
import java.util.*;
public class main{
    static class InsufficientBalanceException extends Exception{
        public InsufficientBalanceException(String message){
            super(message);
        }
    }
    public static void main(String[] args){
        Scanner sc = new Scanner(System.in);
        try{
            double balance = Double.parseDouble(sc.next());
            double withdrawl = Double.parseDouble(sc.next());
            if(withdrawl<=0){
                System.out.println("Invalid withdrawl amount");
            }
            else if(withdrawl>balance){
                throw new InsufficientBalanceException("Insufficient balance");
            }
            else{
                balance =balance-withdrawl;
            }
            System.out.println("Withdrawl successful");
            System.out.println("Remaining balance:"+balance);
        }
        catch(InsufficientBalanceException e){
            System.out.println(e.getMessage());
        }
        finally{
            System.out.println("Transaction completed");
        }
    }
}
```

INPUT

10000
2500

OUTPUT

Withdrawl successful
Remaining balance:7500.0
Transaction completed

Q2. Hospital Patient Registration – User-Defined Exception

CODE

```
import java.util.Scanner;
public class Main{
    static class InvalidAgeException extends Exception{
        public InvalidAgeException(String message){
            super(message);
        }
    }
    static class InvalidTemperatureException extends Exception{
        public InvalidTemperatureException(String message){
            super(message);
        }
    }
    public static void main(String[] args){
        Scanner sc =new Scanner(System.in);
        try{
            String name = sc.next();
            int age = sc.nextInt();
            double temperature = sc.nextDouble();
            if(age<18){
                throw new InvalidAgeException("Invalid Age Exception");
            }
            if(temperature<95.0||temperature>100.0){
                throw new InvalidTemperatureException("InvalidTemperatureException");
            }
            System.out.println("patient registered successfully");
        }
        catch(InvalidAgeException e){
            System.out.println(e.getMessage());
        }
        catch(InvalidTemperatureException e){
            System.out.println("Registration completed");
        }
    }
}
```

INPUT

Meena
12
98.4

OUTPUT

Invalid Age Exception

Q3. E-Commerce Order Processing – Built-in Exceptions

CODE

```
import java.util.Scanner;
import java.util.*;
public class Main{
    public static void main(String args[]){
        Scanner sc= new Scanner(System.in);
        String[] products = {"Laptop","Phone","HeadPhone","Keyboard"};
        double[] prices ={50000,20000,3000,1500};
        try{
            int index = sc.nextInt();
            int quantity = sc.nextInt();
            String product=products[index];
            double price = prices[index];
            if(quantity<=0){
                System.out.println("Invalid quantity");
            }
            else{
                double total = quantity*price;
                System.out.println("Product: "+product);
                System.out.println("Total cost "+total);
            }
        }
        catch(ArrayIndexOutOfBoundsException e){
            System.out.println("Array Index Out of Bound Exception Handled");
        }
        catch(InputMismatchException e){
            System.out.println("Input Mismatch Exception handled");
        }
    }
}
```

INPUT

2
2

OUTPUT

Product: HeadPhone
Total cost 6000.0

Q4. Railway Ticket Booking – Multithreading and Synchronization

— Not Submitted —

Q5. Bank Account – Concurrent Withdrawals

— Not Submitted —
